

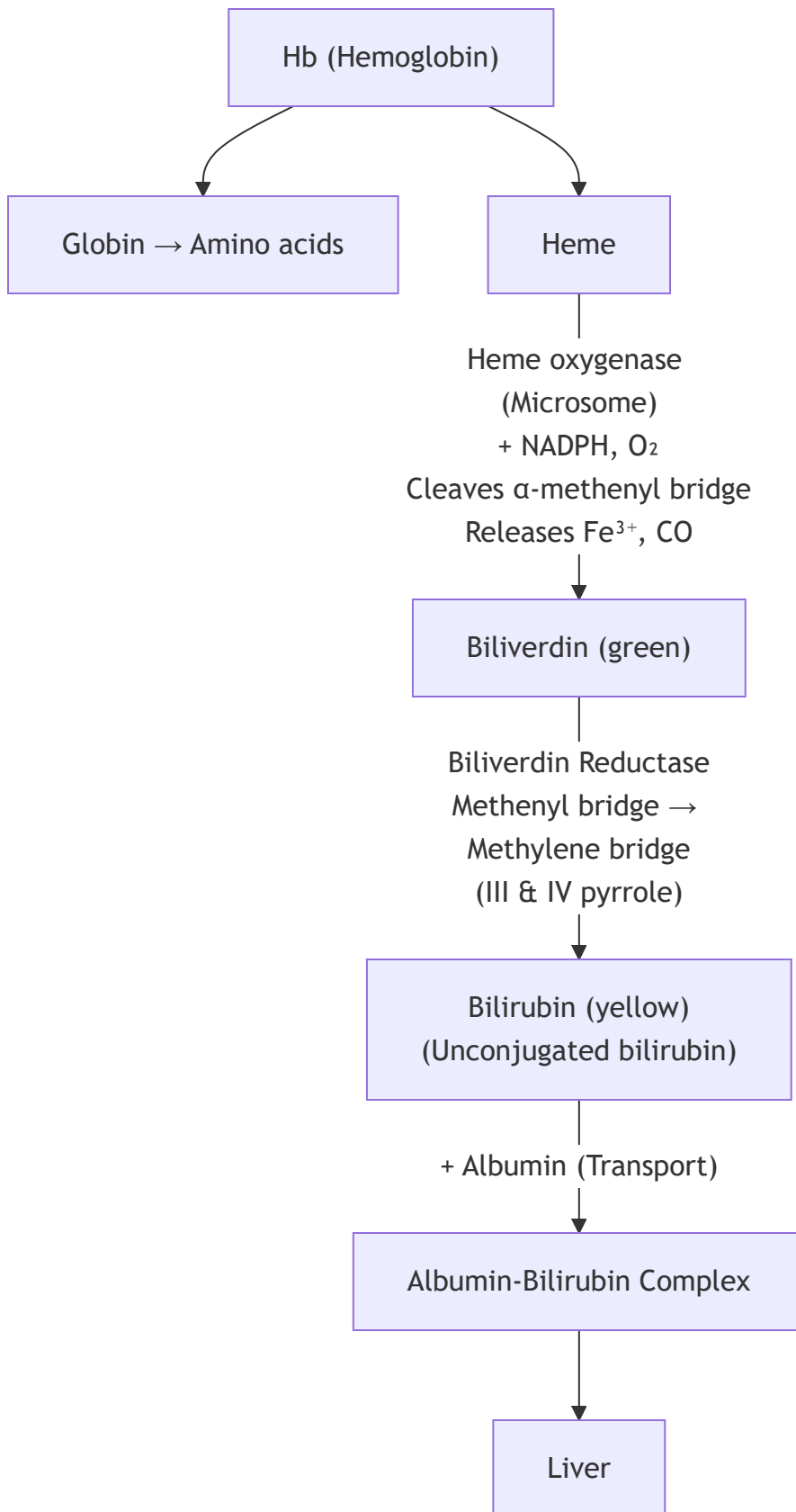
HEME CATABOLISM / BILIRUBIN METABOLISM

Headings

- ▼ HEME CATABOLISM / BILIRUBIN METABOLISM
 - Synthesis & Transport Pathway
 - Albumin Binding Sites
 - Metabolism of Bilirubin
- ▼ BILIRUBIN UPTAKE & CONJUGATION
 - 1. Uptake of Bilirubin in Hepatocyte (Facilitated Diffusion)
 - 2. Conjugation & Hepatic Fate
 - Clinical Summary
- ▼ 3. SECRETION INTO BILE CANALICULI & CONGENITAL HYPERBILIRUBINEMIAS
 - Excretion & Enterohepatic Fate Pathway
 - Quantitative Bilirubin Production
 - Congenital Hyperbilirubinemia
- ▼ TYPES OF HEMOGLOBIN & HEMOGLOBIN DERIVATIVES
 - TYPES OF HEMOGLOBIN
 - ▼ HEMOGLOBIN DERIVATIVES
 - Normal Derivatives
 - Abnormal Derivatives

Site : Macrophages of Reticuloendothelial System (Liver, Spleen, Bone Marrow)

Synthesis & Transport Pathway



Albumin Binding Sites

- **High affinity binding site** : ≈ 25 mg
- **Low affinity site** : Easily detached and diffuses into tissue in diseases

Metabolism of Bilirubin

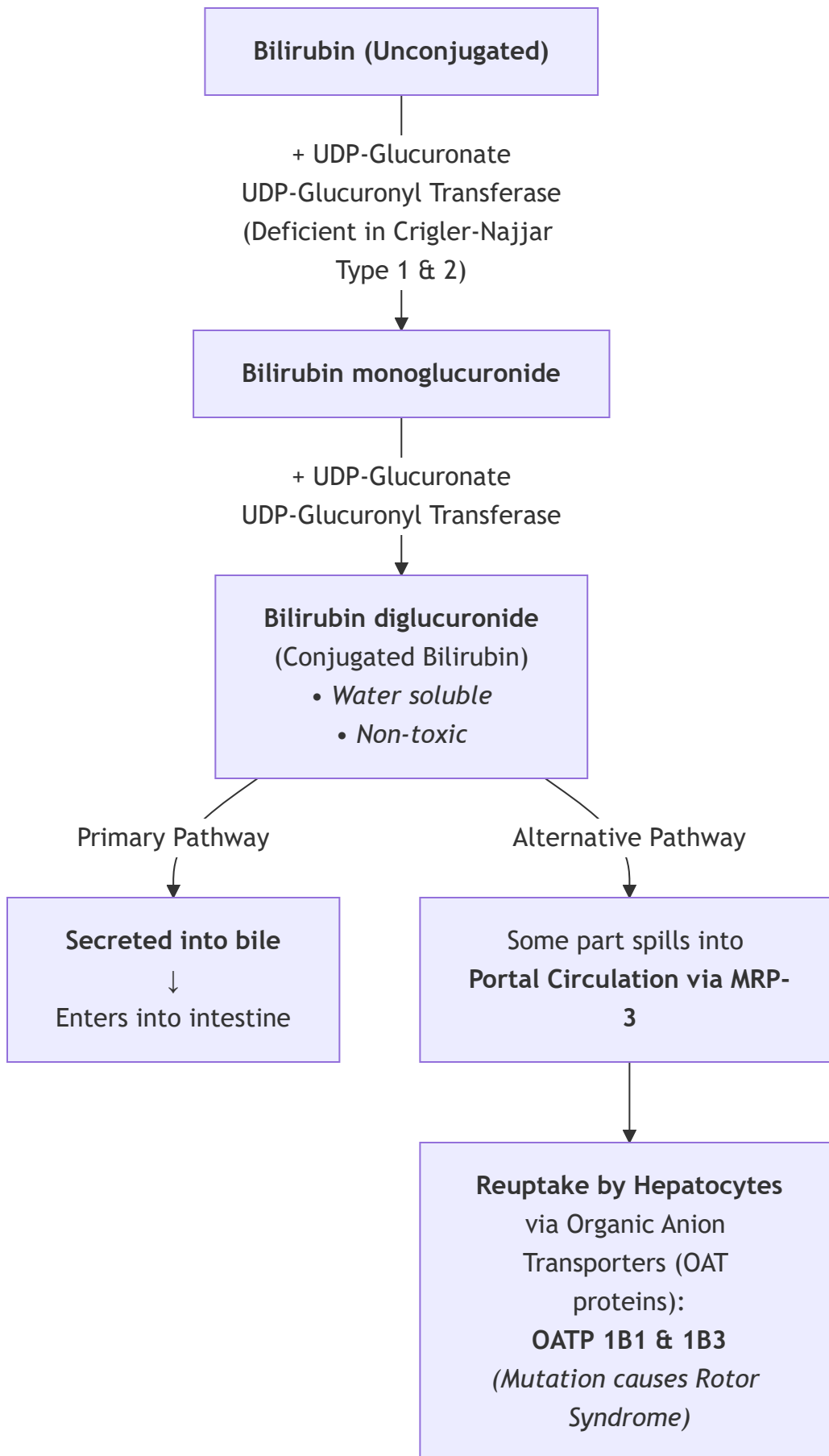
- i. Uptake of bilirubin in Hepatocytes
- ii. Conjugation
- iii. Secretion into bile canaliculi

BILIRUBIN UPTAKE & CONJUGATION

1. Uptake of Bilirubin in Hepatocyte (Facilitated Diffusion)

- **Transport Mechanism** : Albumin-bound bilirubin (Alb-Bil) in the blood dissociates; free bilirubin crosses the sinusoidal membrane into hepatocytes via **facilitated diffusion**.
- **Kinetics** :
 - Large capacity system.
 - **Not a rate-limiting step**, even in pathological states.
- **Intracellular Binding (Ligandin Proteins / Y-Protein)** :
 - Binds bilirubin inside the hepatocyte.
 - **Prevents efflux** of bilirubin back into the bloodstream.
 - Keeps bilirubin in a **solubilized state**.
- **Clinical Correlation** : Defect in uptake is associated with **Gilbert syndrome / disease**.

2. Conjugation & Hepatic Fate



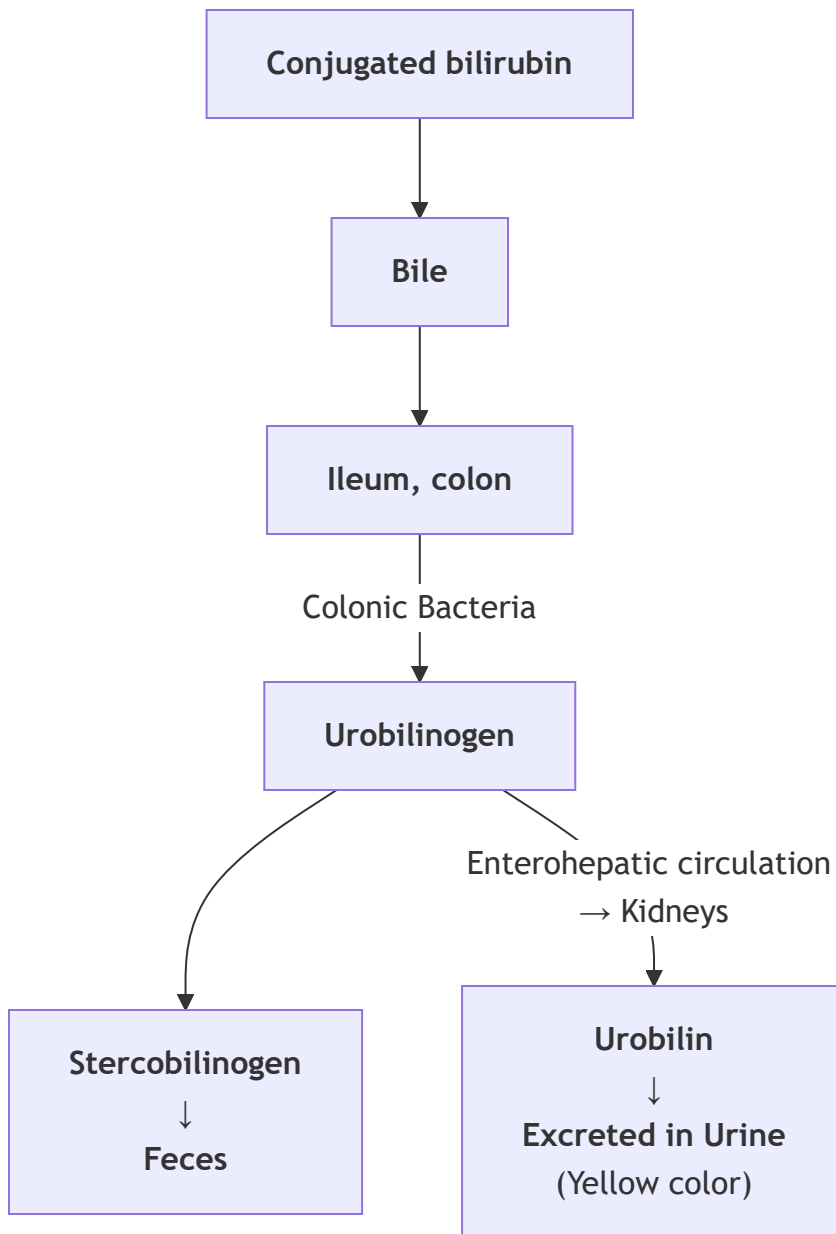
Clinical Summary

- **Crigler-Najjar Syndrome (Types 1 & 2)** : Deficiency of **UDP-Glucuronyl Transferase**, impairing conversion of unconjugated bilirubin to its glucuronide forms.
- **Rotor Syndrome** : Mutation in **OATP 1B1 & OATP 1B3**, impairing hepatic reuptake and storage of conjugated bilirubin.

3. SECRETION INTO BILE CANALICULI & CONGENITAL HYPERBILIRUBINEMIAS

- **Rate Limiting Step** : Active transport via **MRP 2 (Multidrug Resistance-like Protein 2)**.
- **Mutation** : Mutation in MRP 2 leads to **Dubin-Johnson Syndrome**.

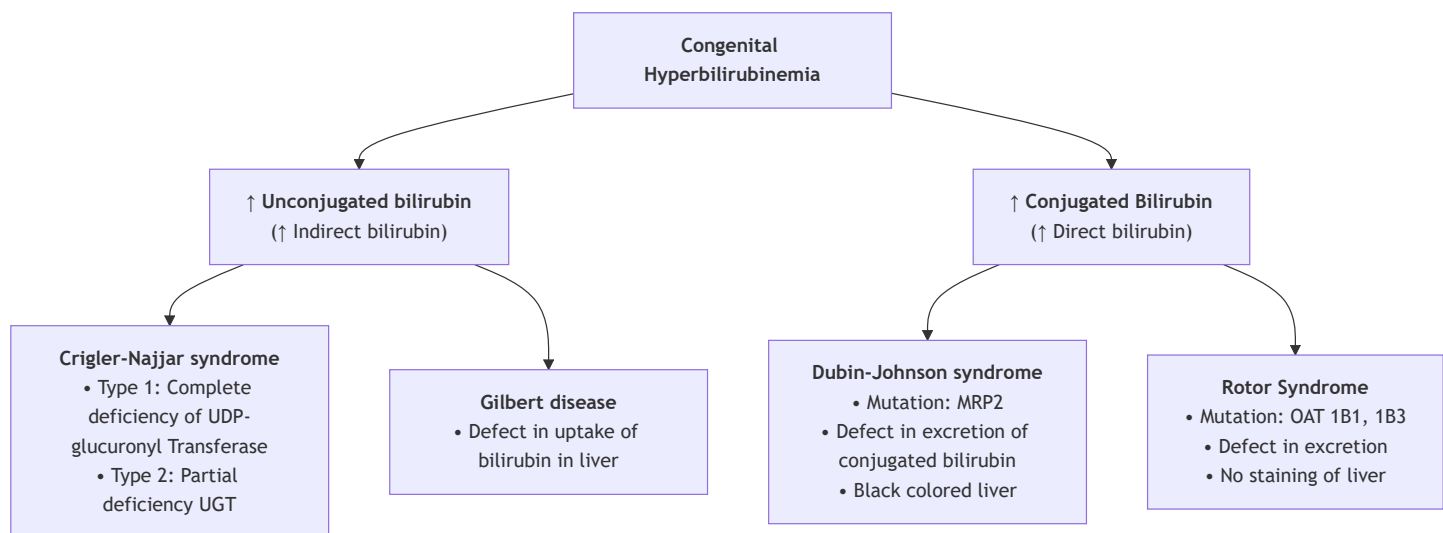
Excretion & Enterohepatic Fate Pathway



Quantitative Bilirubin Production

- 1 gm Hb $\longrightarrow$ 35 mg bilirubin
- 6 gm Hb degraded per day

Congenital Hyperbilirubinemia

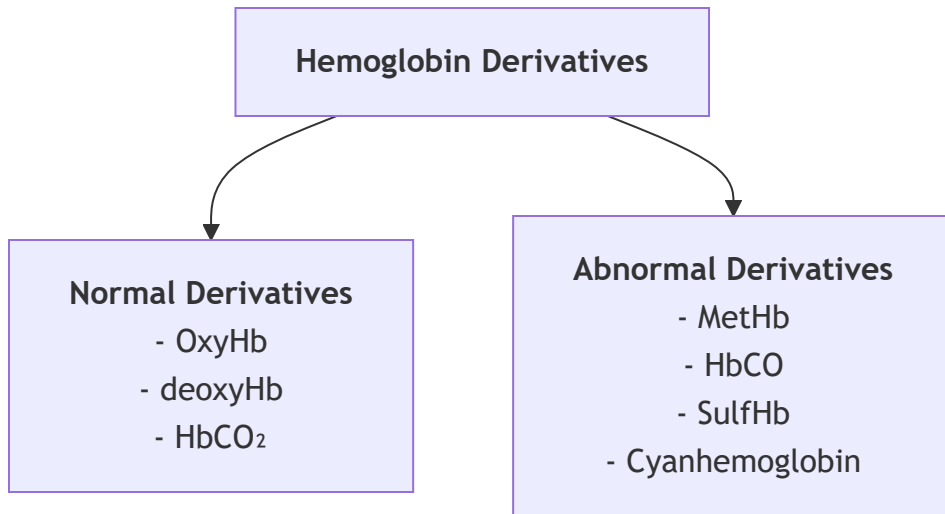


TYPES OF HEMOGLOBIN & HEMOGLOBIN DERIVATIVES

TYPES OF HEMOGLOBIN

Type of Hb	Component Subunit	Description / Significance
Adult Hb (HbA)	$\alpha_2\beta_2$	The major Hb in humans (90%)
HbA ₂	$\alpha_2\delta_2$	Minor component in adult (2 - 5%)
Fetal Hb (HbF)	$\alpha_2\gamma_2$	< 2% in adults
Glycated Hb (HbA _{1c})	$\alpha_2\beta_2$ -Glucose	< 5% ; increases in Diabetes Mellitus (DM)

HEMOGLOBIN DERIVATIVES



Normal Derivatives

- 1. Oxyhemoglobin (OxyHb) :

Hb with O₂

- 2. Deoxyhemoglobin (deoxyHb) :

Hb without O₂

- 3. Carbaminohemoglobin (HbCO₂) :

- CO₂ is non-covalently bound to the globin chain of Hb.
- Transports CO₂ in blood.

Abnormal Derivatives

- 4. Methemoglobin (MetHb) :

- Contains Fe³⁺ instead of Fe²⁺ in the heme group.
- **Unable to transport oxygen.**

- 5. Carboxyhemoglobin / Carbonylhemoglobin (HbCO) :

- CO binds to Fe²⁺ in heme in cases of **CO poisoning** or **smoking**.
- Carbon monoxide has a **much higher affinity** for Fe²⁺ than O₂.

- 6. Sulfhemoglobin & Cyanhemoglobin :

- **Sulfhemoglobin** : Formed in poisoning by **H₂S**.
- **Cyanhemoglobin** : Formed in **cyanide poisoning**.